Capstone Ideation Notes

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Working notes for an immersive-experience capstone (ATLS 4000, Fall 2026). We add to this as the idea takes shape. Nothing here is final.

The common thread: an experience that makes people feel something, uses what they already feel, or works through their senses.

Precedents, each broken into the same building blocks, live in the Immersive Inspiration Log.

For now: keep every thread open. Specifics come later, after research and talking to people.

Team: likely three people.

Approval: whatever we land on has to pass as a CTD capstone. That means a specific audience, a real technical challenge, and a full semester of work for each of three people. The Approval check near the end is for testing ideas later, not now.

Building blocks

Every immersive experience makes the same three choices: input, medium and feeling. Sometimes one is left implicit. A passive projection show like Atelier des Lumières has no input at all. Two more blocks turn an idea into a pitch: a specific audience, and an arc (how people enter, what changes, how they leave).

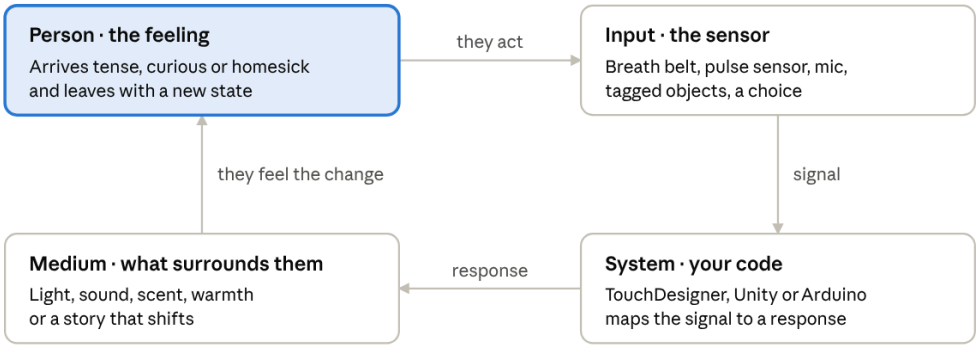
Block	Question it answers	Options to choose from	Seen in
Input	What does the person give the experience?	Breath, heart rate, skin conductance, voice, movement or presence, touch and objects, memories or stories, choices, another person, or nothing (passive)	Pulse Room (heartbeat), Rain Room (presence), Sketch Aquarium (a drawing)
Medium	What surrounds them?	VR headset, projection room, physical set and objects, spatial audio or headphones, scent, temperature, wind, AR on a phone	Carne y Arena (VR plus a sand floor), Dreamachine (light seen through closed eyes)
Feeling	What should they feel or leave with?	Calm, nostalgic warmth, belonging, wonder, empathy, being heard, curiosity	DEEP (calm), Museum of Broken Relationships (shared heartbreak)
Audience	Who is it for, specifically?	Stressed students, older adults with dementia, homesick international students, strangers at Expo, caregivers	Glenner Town Square (dementia day care)

Block	Question it answers	Options to choose from	Seen in
Arc	How long, how many people, and what happens first, middle and last?	One person for 3 minutes, pairs, a walk-through group; a threshold moment before it starts	Carne y Arena's cold waiting room before the headset goes on

Responsive experiences, where input changes what the person sees, hears or smells, are where CTD engineering shows. Passive spectacle is mostly content production.

The loop

The person changes the space, and the space changes the person



Audience and arc wrap around the whole loop: who walks in, and how long they stay.

responsive experience loop · 4 parts

Picking the building blocks means filling each box. The loop is what separates a responsive piece from a projection you just stand inside.

Idea pool

Anything and everything, one line each. Threads stay separate. Nothing gets combined, judged or checked for feasibility yet. Add freely; a half-formed thought is fine.

Travel and place

- Visiting somewhere you've never been, through every sense but sight
- A place that doesn't exist: an imagined city, a dream place
- Going back to a place you used to live
- The feeling of arriving somewhere new: the airport, the train, the first breath of different air
- Traveling through time instead of space: the same spot 100 years ago

- Micro-travel: shrinking down to a forest floor or a drop of water
- Places you can't go: the deep ocean, space, inside the body
- Home, for someone who is far from it
- A journey where the route matters more than the destination
- Bringing a souvenir back from a place that isn't real

Altering your physiological state

- Slowing someone down: breath, heart rate, calm
- Speeding someone up: excitement, a rush, adrenaline
- Awe on purpose: feeling very small
- Rest: sleepiness, a nap space, a reset after overwhelm
- Changing how time feels
- Warmth, weight and pressure on the body
- Sound and vibration you feel in your chest
- Light and colour shifting mood
- Making an invisible state visible: seeing your own stress

Using what people already feel

- The space reads your mood and reflects it back
- Responding to heartbeat, breath, voice, touch or stillness
- Matching how you feel versus shifting it
- A whole crowd's feelings combined into one thing
- Posture, movement or face as input
- Strangers sharing feelings anonymously

Nostalgia and memory

- Universal childhood moments: the school bus, sleepovers, summer evenings
- A specific generation's world: a 2000s bedroom, a 1990s living room
- Smells that bring people back: crayons, sunscreen, rain
- Sounds of the past: dial-up, an ice cream truck
- A memory that fades or changes each time you revisit it
- Nostalgia for something you never lived through
- Old media as the interface: cassettes, Polaroids, VHS
- Memory loss
- A vintage phone that plays strangers' memories (partner)
- Flipping through past eras on an old TV dial (partner)
- A familiar song with pieces missing (partner)

Connection

- Strangers syncing up without talking
- Feeling someone else's heartbeat
- Something two people can only finish together
- Connection across distance
- Connection across time: messages left for the next visitor

- Seeing the world from someone else's perspective
- Loneliness made visible
- Light that travels between two people as they approach (partner)
- Touching together makes the whole piece come alive (partner)

Storytelling

- A story told only through sound
- A story that changes with how you feel
- Visitors' stories become the content
- A story you walk through, room by room
- One object, many stories
- Overlooked stories of a place or community
- A story with no words
- Ordinary objects that each hide a story (partner)

Personal touch

- Leaving something behind that stays in the piece
- Taking something home afterward
- It remembers you when you come back
- Your voice, drawing or handwriting in the work
- Made for one person at a time
- Leaving an anonymous voice memory for the next visitor (partner)

Senses

- Smell as the main medium
- Touch and texture: floors, fabric, temperature, wind
- Taking a sense away: darkness, silence
- Swapping senses: hearing colours, seeing sound
- Taste
- Vibration and haptics
- Sound made by physical behavior alone, like bowls drifting on water (partner)

Identity and time

- Seeing possible past or future versions of yourself (partner)
- A mirror that shows who you almost were (partner)
- What changes about us over time, and what stays

Hidden behaviors in ordinary objects

- Everyday objects with secret responses: a phone, a TV, a music box, a mirror (partner)
- Discovery without instructions
- The physical behavior is the whole artwork, no screen (partner)

Wild cards

- Anything that doesn't fit a thread goes here

Concept sketches

Eight fuller concepts from a possible partner, all sized for a small Expo footprint. They show where threads could go; none of them is a pick.

#	Concept	What happens	Threads	Tech	Footprint
1	The Memory Hotline	Lift a vintage phone's receiver and dial to hear memory fragments. Leave your own anonymous voice memory to grow a shared archive.	Nostalgia, personal touch, connection	Raspberry Pi, rotary dial sensing, microphone, speaker	One pedestal
2	Invisible Threads	Hanging illuminated threads react to proximity. When two people approach together, light travels between them.	Connection	Distance sensors, addressable LEDs, microcontroller, ambient sound	Freestanding frame, about 3 × 3 ft
3	Museum of Ordinary Things	Picking up a familiar object (a house key, a cassette, a toy, a letter) reveals its hidden story through sound, projection or AR. Asks why ordinary objects become extraordinary when they belong to our memories.	Nostalgia, storytelling, hidden behaviors	RFID/NFC tags, object sensors, AR or projection mapping	One tabletop
4	Resonance	Inspired by clinamen. Hands near a shallow pool of floating ceramics shape gentle sound and light, and two people make richer harmonies. A dry version could use suspended resonant objects.	Connection, senses	Capacitive or distance sensors, sound synthesis, LEDs, transducers	Small table

#	Concept	What happens	Threads	Tech	Footprint
5	Channels of Yesterday	An old TV plays dreamlike fragments of past eras. Turning a dial moves through collective memories: old commercials, childhood sounds, forgotten technology.	Nostalgia, travel through time	Rotary encoder, Raspberry Pi, video editing, generative visuals	Small TV stand
6	The Things We Grow	A sculptural garden brightens as people touch its flowers. Touching several at once sends light and sound across the whole garden, like relationships becoming a community.	Connection	Capacitive touch sensors, LEDs, microcontroller, sound	One sculptural table
7	A Mirror of Almosts	An antique-style mirror shows a shifting, abstract version of you. Gestures reveal silhouettes and words about possible past and future selves.	Identity and time	Webcam or depth sensor, TouchDesigner or p5.js, display behind a two-way mirror	Freestanding mirror
8	The Unfinished Music Box	A music box plays a familiar but incomplete melody. Turning a crank or opening compartments restores missing sounds and reveals a story.	Nostalgia, storytelling, hidden behaviors	Magnetic switches, rotary encoders, microcontroller, sound, optional projection	Tabletop sculpture

All eight are object-scale and screen-light. Whether we stay that small is an open question below.

Our project so far

Nothing is decided yet. Fill in a row and change its status as each block settles. When every row says Decided, we have a pitch.

Block	Current answer	Status	Why
Input		Open	

Block	Current answer	Status	Why
Medium		Open	
Feeling		Open	
Audience		Open	
Arc (length, group size, enter and exit)		Open	
Theme	Immersive, sensory, emotionally meaningful	Leaning	Where my interest has been from the start
Team	Likely three people	Leaning	Faculty expect the scope to match three people's work

Approval check

The final proposal is judged on research, feasibility, preparedness and relevance to CTD. Use this once threads start narrowing. For now, nothing has to pass it.

- ☐ A specific audience and a clear problem or experience
- ☐ Improves the world or adds value to an experience
- ☐ Shows technical mastery (a responsive system, not just content)
- ☐ Enough scope for three people, each with a specific role worth 3 credits of spring work
- ☐ Precedent research, including how it differs from Merkaba Meditation, Mycorrhizal and All Things End
- ☐ User interviews or expert conversations (for art pieces: outside experts)
- ☐ Feasibility tests, scariest first
- ☐ Full renderings of the space or experience
- ☐ Production timeline and bill of materials
- ☐ Works at ATLAS Expo: crowded, loud, about 2.5 hours, lots of people walking through
- ☐ Ask the instructor whether user testing with body data needs an IRB determination

Open questions

- ☐ Who are the other two group members, and what role does each take?
- ☐ Headset (VR) or a physical room or installation?
- ☐ Which threads keep pulling at us after a few weeks? (No rush.)
- ☐ Who is the specific audience, and can I reach them for interviews this semester?
- ☐ Which faculty member fits as a sounding board for immersive or biofeedback work?
- ☐ Could the ATLAS B2 black box studio host a test?
- ☐ Visit Meow Wolf Convergence Station in Denver as a field precedent?
- ☐ Small tabletop or pedestal footprint, or a room-scale space?
- ☐ Does it need a screen, or can physical behavior be the whole experience?
- ☐ Ordinary objects with secret behaviors: a direction we want to explore?

- ☐ Should it work for one visitor, or need two people collaborating?

Decision log

Newest first. One line per decision, with the reason.

Date	Decision	Why
Oct 19, 2026	Likely a three-person team	Scope and roles will be planned for three
Oct 19, 2026	Keep every thread open; no combining yet	Specifics come after research and conversations
Oct 19, 2026	Explore immersive experiences as the capstone space	Interest in experiences that create feeling, read what people feel, or use the senses